

- 23** Telomerase is an enzyme that prevents the shortening of telomeres. It is **not** present in most normal cells, but is active in an estimated 85% to 95% of human tumour cells.

Which statement explains the effect of telomerase on human tumour cells?

- A** Telomerase triggers a self-destruct process, known as apoptosis, ending the life of the cell.
- B** Telomerase damages the chromosomes so they become genetically unstable and are unable to replicate and divide.
- C** Telomerase helps human tumour cells avoid senescence, or cell death, which is usually the expected consequence of repeated cell division.
- D** Telomerase enables the human tumour cells to divide more rapidly by reducing the time taken for a complete mitotic cell cycle.